

Day 16: Advanced DAX Iterators (X-Functions)

[DEF] Definition to Remember

Iterator Functions (X-Functions) like SUMX, AVERAGEX, and RANKX evaluate an expression row-by-row over a specified table, creating a Row Context before aggregating the results.

1. SUMX() vs SUM()

- SUM(Table[Column]) only takes a single physical column.

2. Dynamic Ranking: RANKX()

```
Customer Sales Rank =  
RANKX(  
    ALL(Dim_Customer[CustomerName]),  
    [Total Sales],  
    , // Optional value  
    DESC,  
    Dense  
)
```

3. DAX Variables: VAR ... RETURN

Variables improve performance (calculated once and cached) and enhance readability:

4. SELECTEDVALUE()

Returns the single value currently selected in a slicer, or a default fallback if multiple items are selected:

[MUST-WRITE] Must-Master Skills & Interview Points

- Why VAR prevents duplicate query execution in complex branching DAX expressions.
- How RANKX calculates dynamic ranks across table subsets.
- When to use SELECTEDVALUE() for dynamic card titles and parameter cards.

[RECALL] Quick Recall

SUMX/AVERAGEX (Row-by-row iterators) + RANKX (Dynamic ranking) + VAR/RETURN (Cached clean variables)